



SPACE & SATELLITE SOLUTIONS



Company Identity & Objectives

JAVAD GNSS is a global multi-solutions GNSS provider with 200+ patents

- ▶ Customizable solutions
- ▶ Broad product portfolio
- ▶ Open interfacing

200+
Patents

Goals

1. Be the primary PVT provider for all applications
2. Be the industry name in navigation and wireless solutions
3. Fit into everyone's solution

Locations



US Engineering Office
Fort Wright, Kentucky, USA



Global Headquarters
San Jose, California, USA



EU Office
Vilnius, Lithuania

TR-2S LEO | OEM GNSS

TR-2S LEO is a complete GNSS receiver solution with radiation tolerance for space applications. More than just a module, it offers a full package ready to output GNSS positioning and navigation.



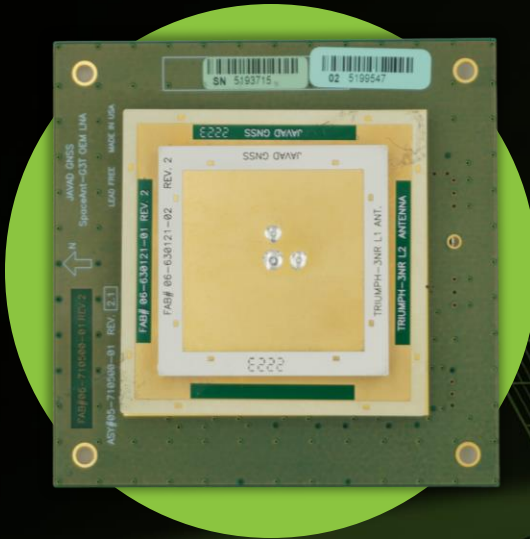
40 x 55 mm / 20 g

- ▶ Spoofing + jamming protection
- ▶ Tracking and acquisition at launch and orbital speeds
- ▶ Capable of 30g acceleration*
- ▶ Radiation tolerant
 - ▶ TID: 25 krad
 - ▶ SEE: No destructive events
- ▶ Fast channel acquisition
- ▶ 4 core processors & 130,000 correlators
- ▶ Advanced multipath reduction



GNSS Antennas | Designed for LEO

- ▶ All GNSS constellations
- ▶ Stable phase center
- ▶ Extended operating temperature
- ▶ Tracking to horizon



SpaceAnt

70 x 70 x 17 mm / 62 g



High-Performance Precision + Accuracy

JAVAD and Fugro provide a complete GNSS hardware + software solution perfect for low Earth orbit (LEO) satellites and applications for autonomous real-time, on-board positioning.

FUGRO
SPACESTAR®

REAL-TIME ACCURATE POSITIONING

SYSTEM ENABLED

**CENTIMETER-LEVEL
ACCURACY IN SPACE**

JAVAD OEM Board Guides ESA Vega Mission

Story Published by

GPS GNSS
POSITIONING
NAVIGATION
TIMING
WORLD

A JAVAD GNSS board is at the heart of the navigation system of the Vega space vehicle, developed by the European Space Agency (ESA) to launch small satellites into Low Earth Orbit.

The JAVAD GNSS board, embedded in the upper stage of the Vega launcher, allows accurate trajectory verification during the entire mission.

It provides great flexibility of the mission at an affordable cost and represents the European solution for space accessibility.

[Read the
Customer Story](#)

JAVAD Supports Student-Led Space Program

The success of Colibri's flights hinged on reliable navigation and control systems. The JAVAD GNSS OEM TR-2S consistently maintained a 1cm fix throughout Colibri's 100-meter flight, despite the harsh vibrations and dynamic flight conditions.



[Read the Customer Story](#)

Image Credit: Gruyère Space Program & volumetrica.studio



JAMMING & SPOOFING

A Growing Threat to Aerospace and Space-Based Infrastructure

Jamming

Jamming disrupts communication by overwhelming the target receiver with noise. It involves transmitting high-powered signals to interfere with the normal operation of navigation systems.

Spoofing

Spoofing aims to deceive the target by mimicking legitimate signals and sending false information. It involves creating and transmitting fake signals that the receiver interprets as authentic.

J-Shield: Patented Protection Built In

J-Shield provides patented protection and detection against signal interference and falsification. Identify and reject falsified signals as soon as they appear.



J-SHIELD

Anti-Spoofing + Anti-Jamming

Custom ASIC at the Core

Fast Acquisition for rapid satellite lock

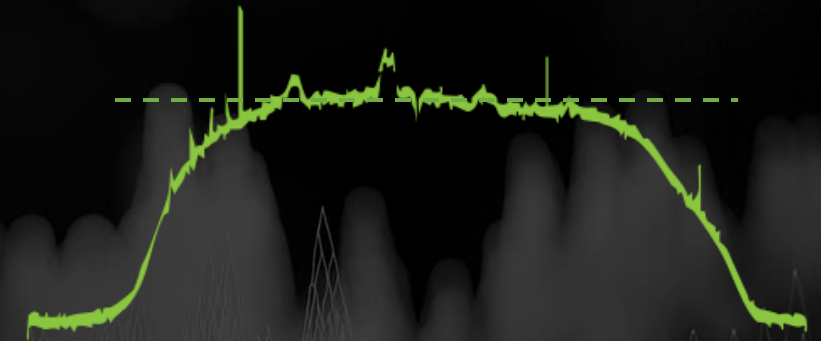
- ▶ 130,000x faster than single correlator search

Continuous Jamming and Spoofing detection with Anti-Jamming Module (AJM)

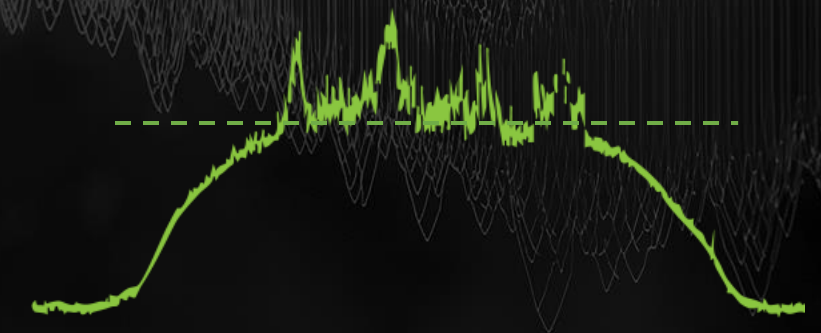
- ▶ Protections built into the core of every JAVAD receiver

Jammed and Spoofed signals are digitized and omitted.

With independent tracking, the remaining valid signals compute PVT.



Typical Spectrum



Jammed Spectrum



END-TO-END CONTROL

From Design to Delivery



Made in the heart of Silicon Valley, California, by our sister company, JAVAD EMS.

JAVAD products are built-in house to withstand the delays and risks of supply chain vulnerabilities. We cover the full product lifecycle, from design to manufacturing and support.

- ✓ **Quality control**
- ✓ **Global support**
- ✓ **Experience**



Certifications

- ▶ EMC Certification
- ▶ Environmental Certification
- ▶ RoHS Certification
- ▶ Declarations of Conformity
- ▶ AS-9100D and ISO 9001 Certified
- ▶ ASR Certified
- ▶ Safe Transport of Goods Reports
- ▶ HSEQ Policy
- ▶ ITAR Registration
- ▶ EU Reach Compliant



9001:2015



13485:2016



JAVAD + Space

Experience the JAVAD GNSS advantage from decades of field-proven experience

- ▶ Centimeter-level positioning for precise orbits
- ▶ Multi-constellation / Multi-frequency tracking
- ▶ Designed & tested for harsh space conditions
 - ▶ Radiation tolerant, low power consumption, MIL-STD shock & vibration
- ▶ Autonomous GNSS tracking & signal processing for LEO constellations
- ▶ RTK & PPP support
- ▶ Real-time positioning for observation, defense, and commercial satellites



**SIMPLE INTEGRATION,
WORLD CLASS
SUPPORT**

JAVAD GNSS Advantages







ONBOARD